

PROJECT SPECIFICATIONS • TEAM ID: UNM293SMARTSTUDYAI

ComicCraft: AI Comic Story Creator using Gemini Models

An end-to-end intelligent platform generating dynamic comic storyboards, panel breakdowns, descriptive visual prompts, and compelling character dialogues using Google Gemini Multimodal APIs.

Project Description

ComicCraft harnesses the multimodal and advanced reasoning capabilities of Google Gemini Models to provide an intelligent, interactive comic creation and storytelling workflow. The platform features an AI Narrative & Script Engine that transforms user story premises into multi-panel comic book layouts, a Visual Scene Prompt Generator that creates detailed visual composition descriptions for panel illustrators or image generation pipelines, and a Character Dialogue & Sound-FX Engine that produces character-consistent speech bubbles, dramatic captions, and expressive comic onomatopoeia tailored to target genres and demographics.

Utilizing the Google Gemini API (including Gemini 1.5 Pro and Gemini 1.5 Flash models), ComicCraft processes complex story ideas, character sheets, and plot outlines to deliver structured, production-ready comic scripts. Built with Flask on the backend and deployed seamlessly via Ngrok for live demonstrations, the platform ensures an intuitive, responsive user experience. With secure API key orchestration via .env and robust input validation, ComicCraft empowers independent writers, comic creators, illustrators, and educators to bring imaginative comic universes to life with confidence.

Use Case Scenarios

Scenario 1: Sci-Fi Adventure Graphic Novel Episode

A writer outlines a cyberpunk detective premise set in Neo-Tokyo. ComicCraft analyzes the world-building parameters and generates a 4-panel comic storyboard complete with establishing camera angles (wide aerial shot vs. high-contrast Dutch tilt), dynamic character dialogue, and vivid lighting notes tailored for graphic novel production.

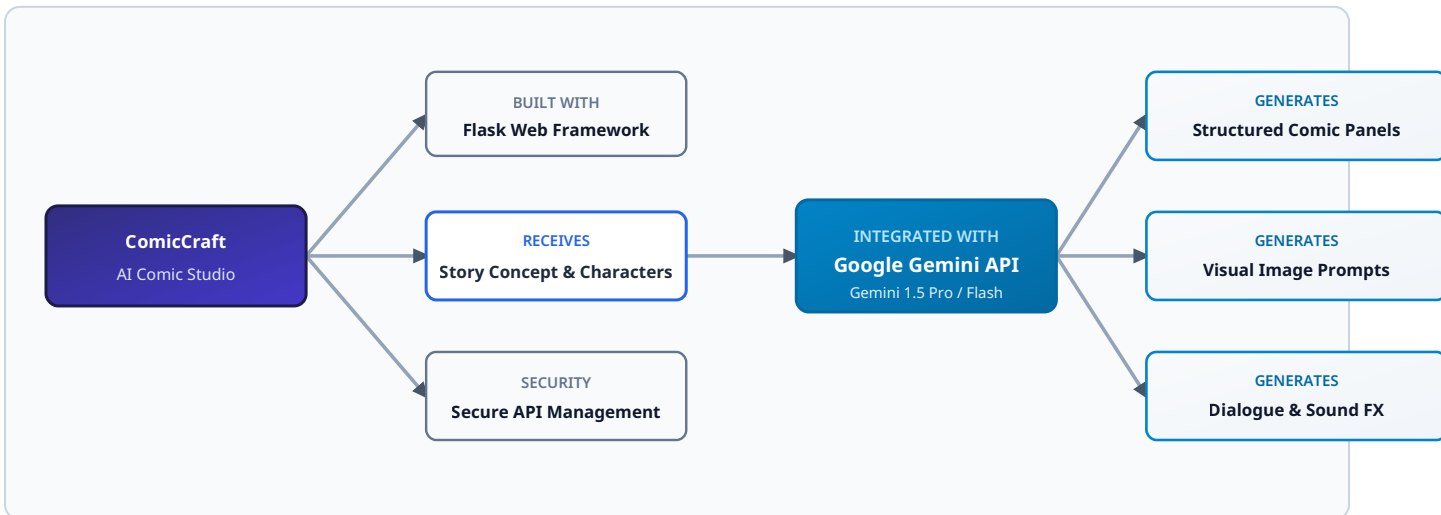
Scenario 2: Educational STEM Comic for Young Readers

A teacher inputs a science topic explaining photosynthesis and solar cells. ComicCraft structures an engaging, child-friendly narrative featuring animated bot companions, breaking down complex concepts into episodic 3-panel strips with humorous dialogue, visual metaphors, and sound effects to maximize classroom engagement.

Scenario 3: Manga Action Scene with Character Consistency

A manga enthusiast submits existing character profiles and a climactic martial arts duel synopsis. ComicCraft generates sequential action beats with pacing guidelines, sound effects (SFX) typography recommendations, emotional close-ups, and dramatic panel border breaks to maximize visual tension and impact.

Technical Architecture



Description: ComicCraft utilizes a modular three-tier architecture to deliver rapid, AI-driven comic book creation. The frontend provides a responsive user interface built with HTML5, CSS3, and JavaScript, while the Flask-based backend orchestrates data validation, template rendering, and prompt construction. It interfaces with Google's Gemini Multimodal models to generate sequential panel scripts, shot composition prompts, and character dialogues, with secure local deployment exposed publicly through Ngrok.

(Note: If you are using a database such as SQLite or MongoDB in your project for saving stories and user comics, include the ER Diagram along with table specifications.)

Pre-requisites

Flask Framework Knowledge:	Flask Official Documentation (Routing, Jinja2, REST APIs)
Gemini API Familiarity:	Google AI Studio & Gemini Python SDK (google-genai)
HTML, CSS, and JavaScript Skills:	Modern Responsive Web Design & DOM Manipulation
Python Programming Proficiency:	Python 3.10+ / Virtual Environments / Requests
Version Control with Git:	GitHub Repository Management & Collaboration
Development Environment Setup:	VS Code, Virtualenv & Python-dotenv Configuration
Ngrok for Public Deployment:	Ngrok Tunneling Documentation & Port Forwarding

Project Workflow

Milestone 1

Model Selection and Architecture

- **Activity 1.1:** Obtain and configure Google Gemini API key securely in project environment variables (.env).
- **Activity 1.2:** Research and select suitable Gemini models (Gemini 1.5 Pro for rich narrative generation; Gemini 1.5 Flash for rapid panel iteration).
- **Activity 1.3:** Define the system architecture, detailing client-server interactions, AI prompt structure, and comic storyboard schemas.
- **Activity 1.4:** Set up the project virtual environment, installing dependencies for Flask, google-genai, and python-dotenv.

Milestone 2**Core Functionalities Development**

- **Activity 2.1:** Develop core comic engines: Comic Script Generator, Visual Panel Prompt Styler, and Character Dialogue Generator.
- **Activity 2.2:** Implement the Flask backend to orchestrate requests, construct multi-turn system prompts, and validate Gemini API responses.

Milestone 3**app.py Development**

- **Activity 3.1:** Write the main application logic in `app.py`, creating web routes for comic creation, character presets, and storyboard rendering.

Milestone 4**Frontend Development**

- **Activity 4.1:** Design a comic-book inspired interface with HTML, CSS, and JS, featuring panel grids, speech bubbles, and style selector cards.
- **Activity 4.2:** Render dynamic templates using Flask Jinja2 to display multi-panel comic previews, downloadable scripts, and generated prompts.

Milestone 5**Deployment**

- **Activity 5.1:** Prepare application for local deployment by validating `requirements.txt` and testing all Gemini API pipelines.
- **Activity 5.2:** Deploy application publicly using Ngrok tunnel to expose local Flask server over a secure HTTPS URL for live evaluation.

Milestone 6**Conclusion**

- **Activity 6.1:** Perform end-to-end user testing across multiple comic genres (Manga, Western Superhero, Graphic Novel, Webtoon).
- **Activity 6.2:** Finalize documentation, update GitHub repository, and prepare project demonstration showcase.

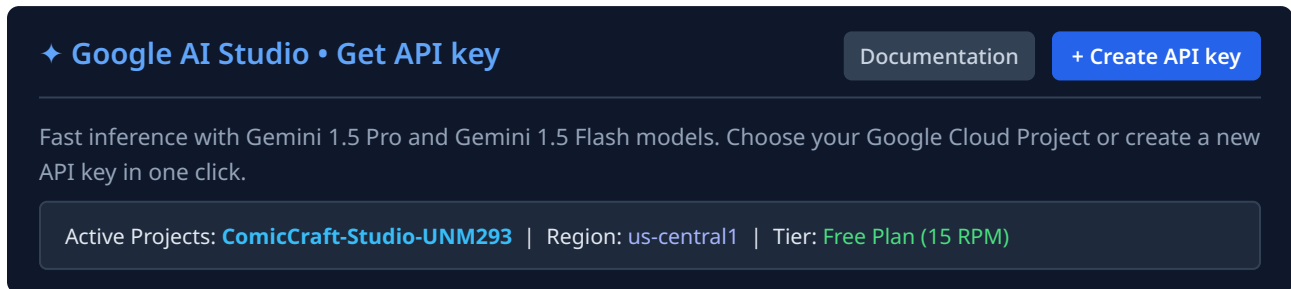
Milestone 1: Model Selection and Architecture — Detailed Setup Guide

In this milestone, we focus on configuring the generative AI model pipeline using Google's Gemini models for our comic storyboard and script creation needs. This involves researching model capabilities, establishing secure API credentialing via Google AI Studio, and ensuring the application architecture supports responsive comic generation.

Activity 1.1: Generate a Google Gemini API Key

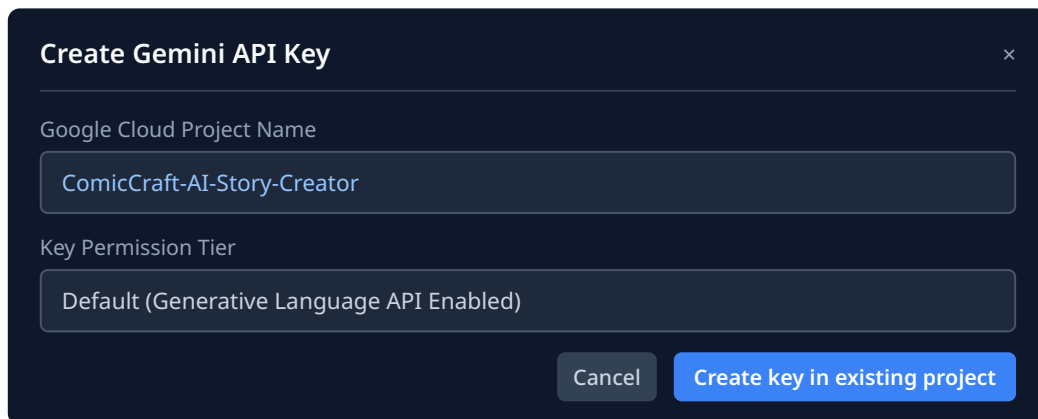
Before the application can communicate with Google Gemini LLMs, you must create an API key via Google AI Studio. Follow the step-by-step instructions below:

- **Step 1 — Access Google AI Studio:** Visit aistudio.google.com and sign in with your Google account.



- **Step 2 — Navigate to API Keys:** Click on the "Get API key" tab in the left navigation sidebar or directly click the blue "Create API key" button.

- **Step 3 — Select or Create a Project:** In the modal popup, select an existing Google Cloud Project or choose "Create API key in new project".



- **Step 4 — Copy and Secure your Gemini Key:** Copy the generated API string (e.g. AIzaSy...) to your clipboard immediately.
- **Step 5 — Configure Environment File:** Create a .env file in your project root directory and add your key:

```
GEMINI_API_KEY="AIzaSyYourGeneratedGeminiKeyGoesHere" FLASK_ENV="development"
FLASK_PORT=5000
```

- **Step 6 — Verify Key Initialization in app.py:**

```
import os from dotenv import load_dotenv from google import genai # Load credentials from
.env load_dotenv() api_key = os.getenv("GEMINI_API_KEY") # Initialize Google Gemini Client
client = genai.Client(api_key=api_key)
```

Important Security Notice: Never commit your .env file to public repositories. Ensure .env and __pycache__ are added to your .gitignore file to prevent accidental exposure of your Gemini API credentials.